PHIVOLCS Special Report – The 10 February 10 2017 Ms 6.7 Surigao Del Norte Earthquake

Epcenter/s: At 9.80°N, 125.35°E

Intensity VIII

Reported Earthquake Intensity	Province	City/Municipality
VIII (very destructive)	Surigao del Norte	Along the surface rupture area (Brgy. Ipil in Surigao City and Barangays Poblacion, Honrado and Macopa in San Francisco)

Places	Coordinates	Findings (from the Method)
Ipil, Surigao City	9.7756° N, 125.4398° E	
Poblacion, San Fracisco	9.7762° N, 125.4248° E	
Honrado, San Francisco	9.7607° N, 125.4261° E	
Macopa, San Francisco	9.7513° N, 125.4261° E	

Intensity-Based Epicenter Estimate Report

1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): 9.761271° N, 125.396756° E

Epicenter 1σ interval: (H ± 0.005551°, K ± 0.069381°)

Distance to PHIVOLCS Measured Epicenter: 6.693 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	0.045366	4.979
P2	0.031770	3.493
P3	0.029350	3.216
P4	0.030992	3.402

Minimum R: 0.029350° (~3.216 km)

Maximum R: 0.045366° (~4.979 km)

Mean R_prox: 0.034370°

Radius std: 0.006409°

2) Recalculated Epicenter (After Outlier Removal)

Epicenter (H, K): 9.761271° N, 125.396756° E

Epicenter 1σ interval: (H ± 0.005551°, K ± 0.069381°)

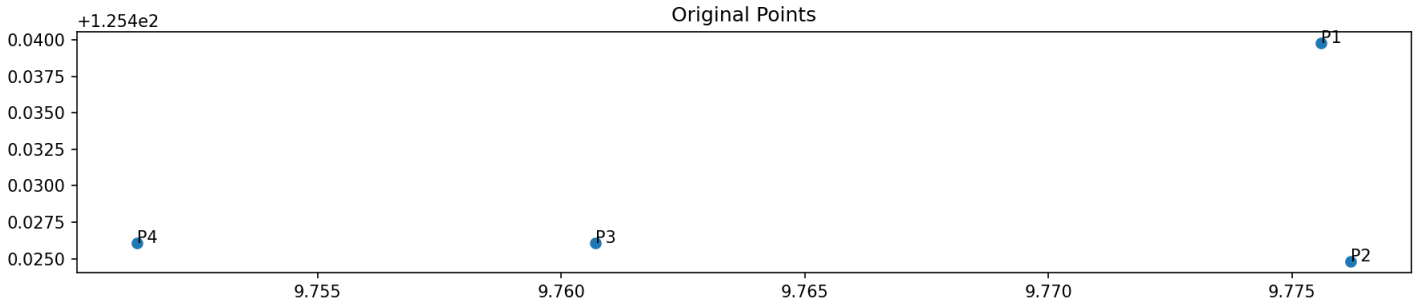
Distance to PHIVOLCS Measured Epicenter: 6.693 km

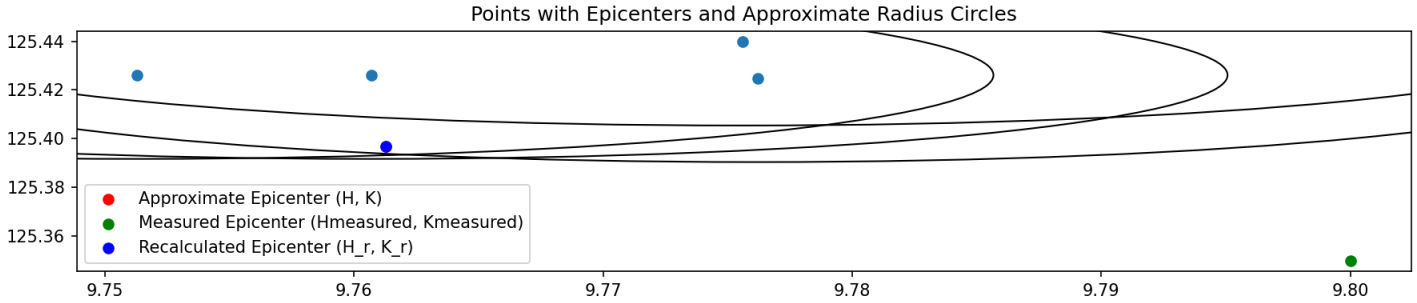
Radii to (H, K)

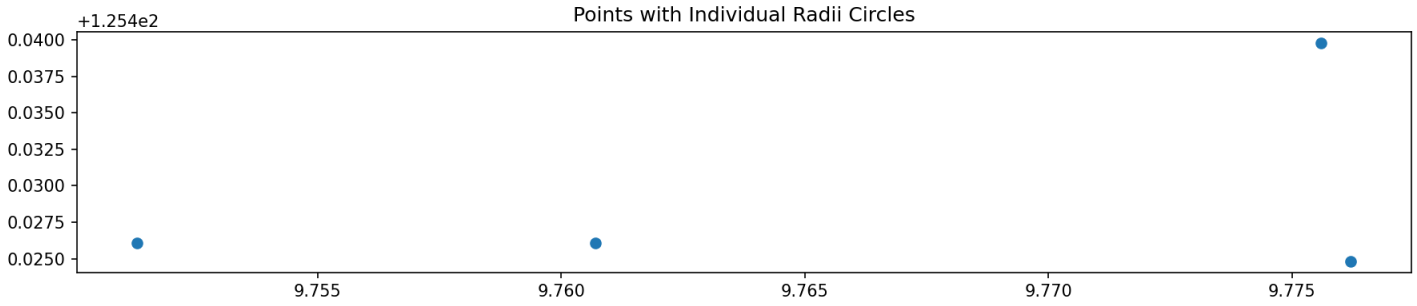
Point	Radius (deg)	Radius (km)
P1	0.045366	4.979
P2	0.031770	3.493
P3	0.029350	3.216
P4	0.030992	3.402

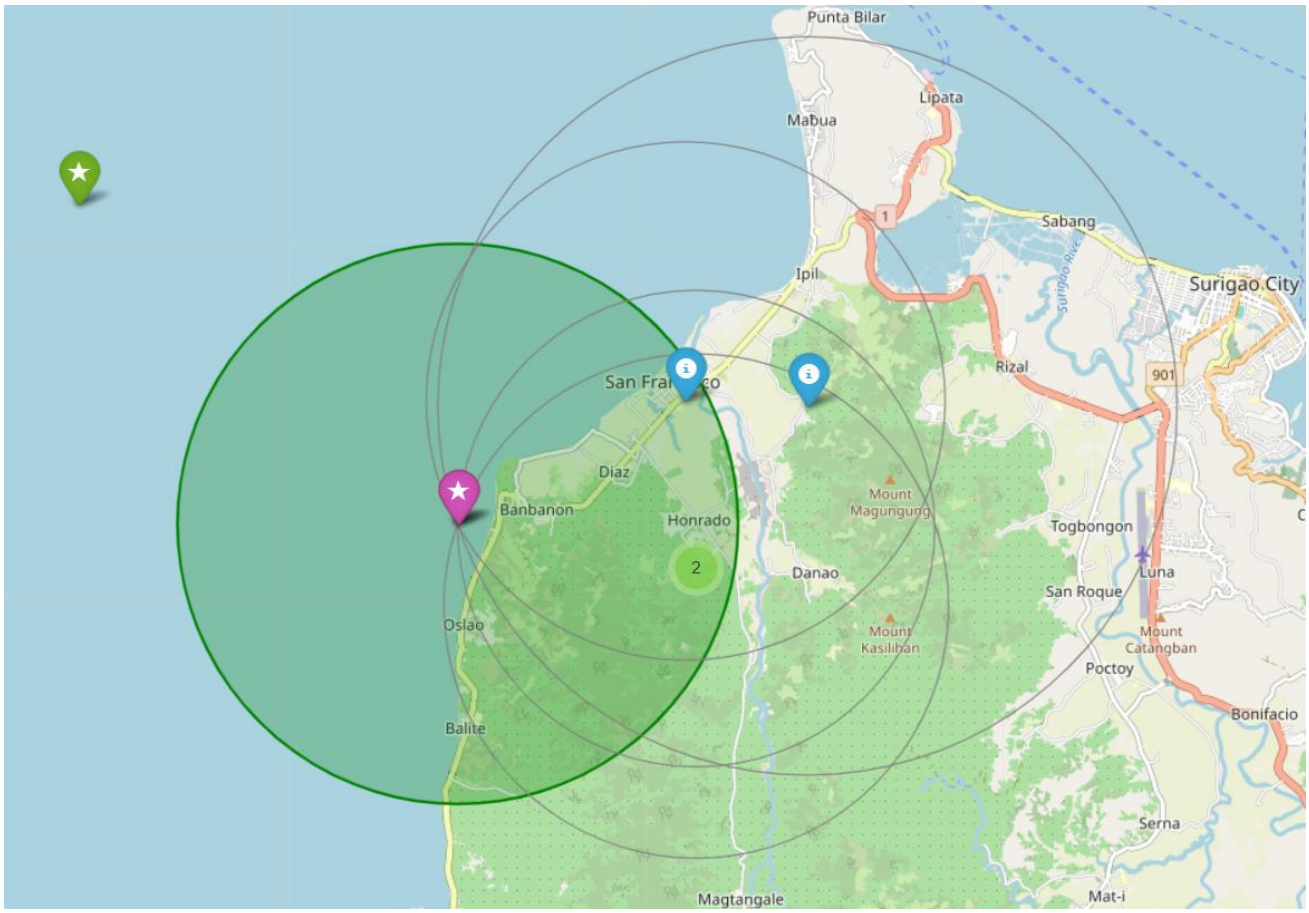
Minimum R: 0.029350° (~3.216 km)

Maximum R: 0.045366° (~4.979 km)


Original Points
+1.254e2
0.0400
0.0375
0.0350
0.0325
0.0300
0.0275
0.0250
P1
P2
P3
P4
9.755
9.760
9.765
9.770
9.775


Points with Epicenters and Approximate Radius Circles
125.44
125.42
125.40
125.38
125.36
Approximate Epicenter (H, K)
Measured Epicenter (Hmeasured, Kmeasured)
Recalculated Epicenter (H_r, K_r)
9.75
9.76
9.77
9.78
9.79
9.80


Points with Individual Radii Circles
+1.254e2
0.0400
0.0375
0.0350
0.0325
0.0300
0.0275
0.0250
9.755
9.760
9.765
9.770
9.775


Punta Bilar
Lipata
Mabua
Sabang
Ipil
Surigao City
Rizal
901
Diaz
Banbanon
Honrado
Mount Magungung
Togbongon
Danao
Luna
San Roque
Oslao
Mount Kasilihan
Mount Catangban
Poctoy
Bonifacio
Balite
Serna
Magtangale
Mat-i

Intensity VI

VI (very strong)	Surigao del Norte	Malimono, Mainit, Placer, Sison and Taganaan

Places	Coordinates	Findings (from the Method)
Malimono, Surigao de Norte	9.5938° N, 125.4234° E	
Mainit, Surigao del Norte	9.5566° N, 125.5001° E	
Placer, Surigao del Norte	9.6348° N, 125.5548° E	
Sison, Surigao del Norte	9.6426° N, 125.4891° E	
Taganaan, Surigao del Norte	9.6887° N, 125.5767° E	

Intensity-Based Epicenter Estimate Report

1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): 9.594538° N, 125.508760° E

Epicenter 1σ interval: (H ± 0.161976°, K ± 0.127575°)

Distance to PHIVOLCS Measured Epicenter: 28.719 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	0.085363	9.359
P2	0.038914	4.324
P3	0.061161	6.747
P4	0.051927	5.762
P5	0.116113	12.849

Minimum R: 0.038914° (~4.324 km)

Maximum R: 0.116113° (~12.849 km)

Mean R_prox: 0.070696°

Radius std: 0.027317°

2) Recalculated Epicenter (After Outlier Removal)

Epicenter (H, K): 9.594538° N, 125.508760° E

Epicenter 1σ interval: (H ± 0.161976°, K ± 0.127575°)

Distance to PHIVOLCS Measured Epicenter: 28.719 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	0.085363	9.359
P2	0.038914	4.324
P3	0.061161	6.747
P4	0.051927	5.762
P5	0.116113	12.849

Intensity V

V (strong)	Surigao del Norte	Tubod, Bacuag, Alegria, Gigaquit and Claver
	Southern Leyte	Pintuyan, Libagon, St. Bernard, Liloan, San Francisco and San Ricardo
	Dinagat Islands	San Jose, Libjo and Cagdianao

Places	Coordinates	Findings (from the Method)
Tubod, Surigao del Norte	9.5784° N, 125.5548° E	
Bacuag, Surigao del Norte	9.5788° N, 125.6314° E	
Alegria, Surigao del Norte	9.4982° N, 125.5986° E	
Gigaquit, Surigao del Norte	9.4854° N, 125.7079° E	
Claver, Surigao del Norte	9.5046° N, 125.7844° E	
Pintuyan, Southern Leyte	9.9951° N, 125.2258° E	
Libagon, Southern Leyte	10.3674° N, 125.0608° E	
St. Bernard, Southern Leyte	10.3223° N, 125.1198° E	
Liloan, Southern Leyte	10.1144° N, 125.1819° E	
San Francisco, Southern Leyte	10.0790° N, 125.1599° E	
San Ricardo, Southern Leyte	9.9952° N, 125.2643° E	
San Jose, Dinagat Islands	10.0196° N, 125.5760° E	
Libjo, Dinagat Islands	10.1877° N, 125.5877° E	
Cagdianao, Dinagat Islands	10.1065° N, 125.6314° E	

Intensity-Based Epicenter Estimate Report

1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): 10.226716° N, 125.882219° E
Epicenter 1σ interval: (H ± 5.150918°, K ± 6.823464°)
Distance to PHIVOLCS Measured Epicenter: 75.152 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	0.726304	80.518
P2	0.694770	77.106
P3	0.781777	86.762
P4	0.761536	84.614
P5	0.728711	81.008
P6	0.696083	76.333
P7	0.833379	91.218
P8	0.768387	84.092
P9	0.709268	77.659
P10	0.737268	80.749
P11	0.659866	72.376
P12	0.369685	40.669
P13	0.297092	32.521
P14	0.278140	30.534

Minimum R: 0.278140° (~30.534 km)
Maximum R: 0.833379° (~91.218 km)
Mean R_prox: 0.645876°
Radius std: 0.178482°

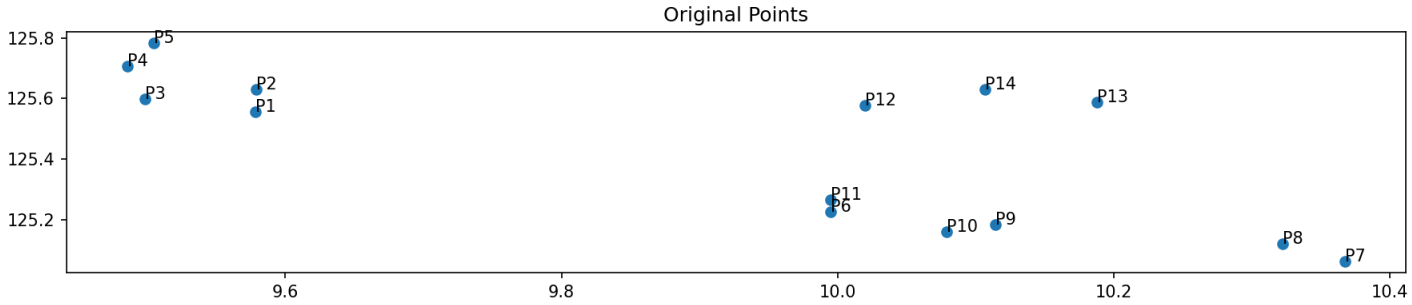
Radii to (H, K)

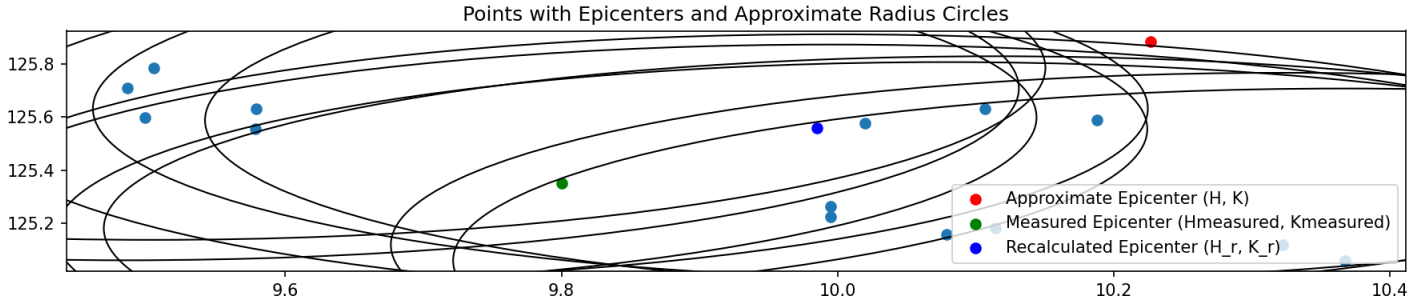
Point	Radius (deg)	Radius (km)
P5	0.531212	58.915
P7	0.627292	69.064
P8	0.553029	60.890
P12	0.038629	4.282
P13	0.204549	22.738
P14	0.141570	15.678

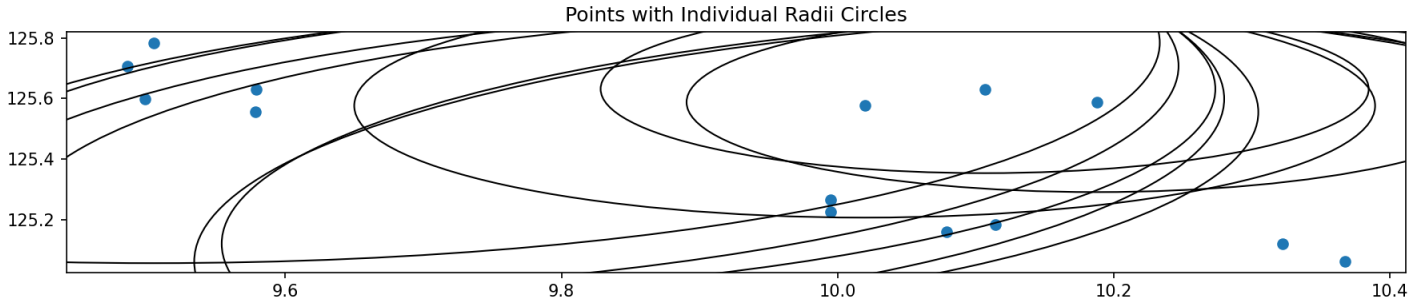
Minimum R: 0.038629° (~4.282 km)
Maximum R: 0.627292° (~69.064 km)
Mean R_prox: 0.349380°
Radius std: 0.228217°

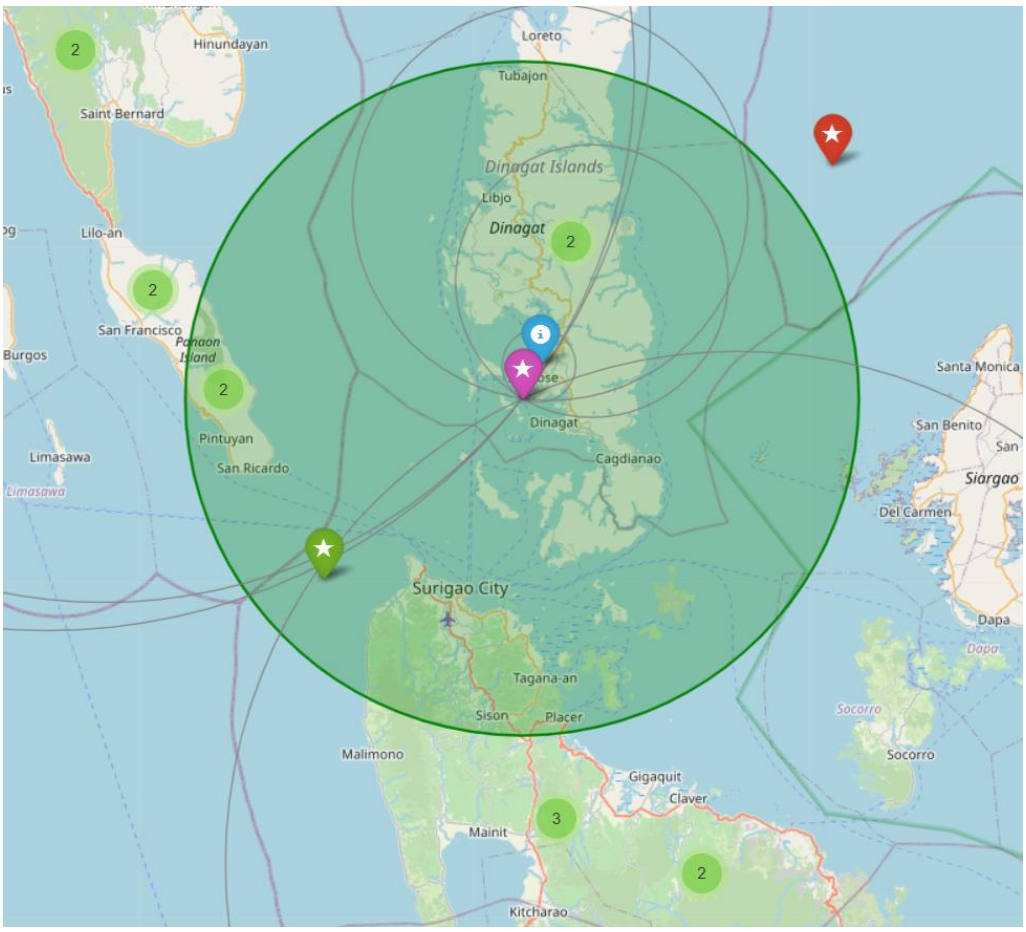
2) Recalculated Epicenter (After Outlier Removal)

Epicenter (H, K): 9.985279° N, 125.558272° E
Epicenter 1σ interval: (H ± 1.117680°, K ± 1.683997°)
Distance to PHIVOLCS Measured Epicenter: 30.740 km


Original Points
P5
P4
P3
P2
P1
P12
P14
P13
P11
P6
P10
P9
P8
P7
125.8
125.6
125.4
125.2
9.6
9.8
10.0
10.2
10.4


Points with Epicenters and Approximate Radius Circles
Approximate Epicenter (H, K)
Measured Epicenter (Hmeasured, Kmeasured)
Recalculated Epicenter (H_r, K_r)
125.8
125.6
125.4
125.2
9.6
9.8
10.0
10.2
10.4


Points with Individual Radii Circles
125.8
125.6
125.4
125.2
9.6
9.8
10.0
10.2
10.4


Hinundayan
Saint Bernard
Loreto
Tubajon
Dinagat Islands
Libjo
Dinagat
Lilo-an
San Francisco
Panaon Island
Burgos
Pintuyan
San Ricardo
Limasawa
Cagdianao
Santa Monica
San Benito
Siargao
Del Carmen
Surigao City
Dapa
Tagana-an
Sison
Placer
Malimono
Socorro
Gigaquit
Claver
Mainit
Kitcharao

Intensity IV

IV (moderately strong)	Southern Leyte	Hinunangan, San Juan, Sogod, Macrohon and Maasin City
	Leyte	Abuyog and Mayorga
	Agusan del Norte	Butuan City and Cabadbaran

Places	Coordinates	Findings (from the Method)
Hinunangan, Southern Leyte	10.3946° N, 125.1489° E	
San Juan, Southern Leyte	10.2638° N, 125.2039° E	
Sogod, Southern Leyte	10.4493° N, 125.0168° E	
Macrohon, Southern Leyte	10.0840° N, 124.9617° E	
Maasin City, Southern Leyte	10.1327° N, 124.8408° E	
Abuyog, Leyte	10.6735° N, 125.0608° E	
Mayorga, Leyte	10.8850° N, 124.9837° E	
Butuan City, Agusan del Norte	8.9475° N, 125.5406° E	
Cabadbaran, Agusan del Norte	9.1202° N, 125.5352° E	

Intensity-Based Epicenter Estimate Report

1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): 9.162475° N, 122.294206° E

Epicenter 1σ interval: (H ± 2.868651°, K ± 9.172126°)

Distance to PHIVOLCS Measured Epicenter: 342.560 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	3.109246	341.497
P2	3.111147	341.605
P3	3.011385	330.851
P4	2.822186	309.867
P5	2.725156	299.281
P6	3.152339	346.488
P7	3.193818	351.291
P8	3.253504	357.284
P9	3.241270	355.835

Minimum R: 2.725156° (~299.281 km)

Maximum R: 3.253504° (~357.284 km)

Mean R_prox: 3.068894°

Radius std: 0.173879°

2) Recalculated Epicenter (After Outlier Removal)

Epicenter (H, K): 9.399025° N, 123.061508° E

Epicenter 1σ interval: (H ± 1.904731°, K ± 5.973798°)

Distance to PHIVOLCS Measured Epicenter: 254.836 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P2	2.310342	253.655
P3	2.219514	243.935
P4	2.019881	221.736
P5	1.924619	211.359
P7	2.429597	267.519
P8	2.519875	276.729

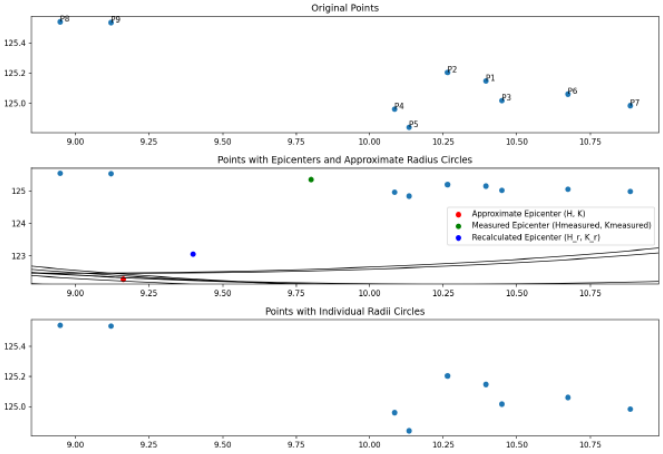
Minimum R: 1.924619° (~211.359 km)

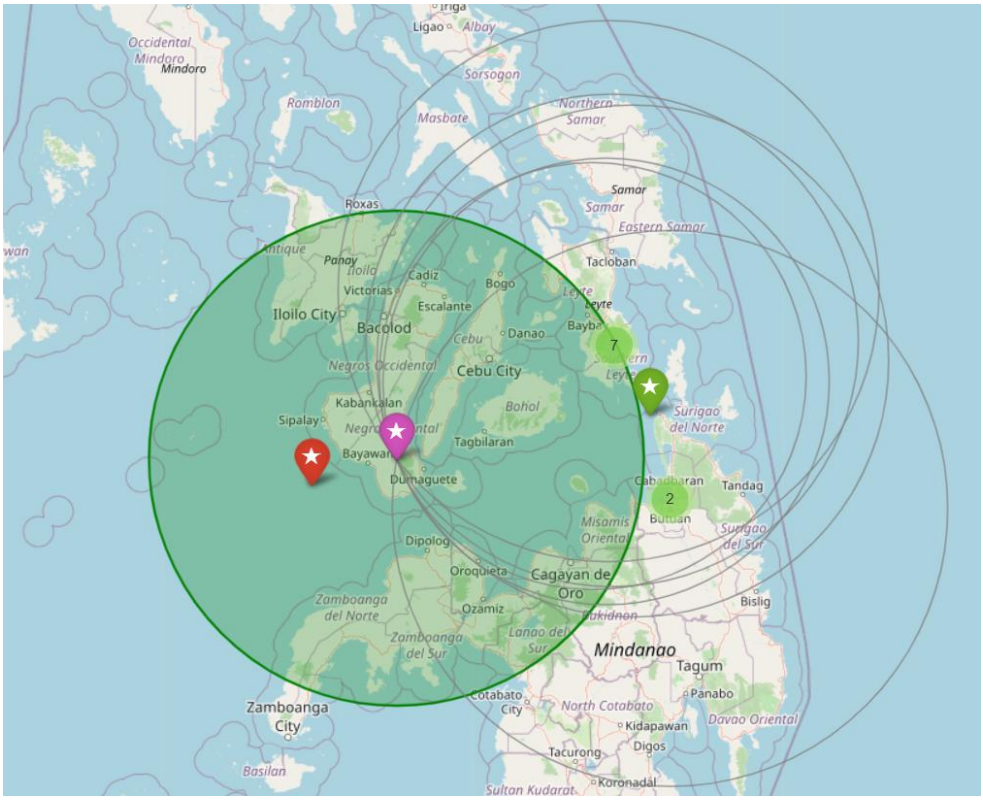
Maximum R: 2.519875° (~276.729 km)

Mean R_prox: 2.237305°

Radius std: 0.211155°

Figure: Triplet Geometry and Radii





Intensity III

Places	Coordinates	Findings (from the Method)
Mambajao, Camiguin	9.2215° N, 124.7188° E	
Tacloban City, Leyte	11.2224° N, 124.9922° E	
Ormoc City, Leyte	11.0384° N, 124.6193° E	
Palo, Leyte	11.1585° N, 124.9919° E	
Pastrana, Leyte	11.1197° N, 124.8845° E	
Tolosa, Leyte	11.0388° N, 125.0223° E	
Catbalogan City, Leyte	11.7752° N, 124.8828° E	
Bislig City, Surigao del Sur	8.1836° N, 126.3567° E	
Cagayan de Oro City, Misamis Oriental	8.4803° N, 124.6498° E	
Gingoog City, Misamis Oriental	8.8145° N, 125.1041° E	
Dapa, Surigao del Norte	9.7892° N, 126.0676° E	
Iligan City, Lanao del Norte	8.2271° N, 124.4642° E	

Intensity-Based Epicenter Estimate Report

1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): 10.306391° N, 123.259784° E

Epicenter 1σ interval: (H ± 2.080113°, K ± 16.775348°)

Distance to PHIVOLCS Measured Epicenter: 235.677 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	1.818163	200.288
P2	1.959677	214.914
P3	1.544059	169.393
P4	1.930366	211.627
P5	1.816913	199.219
P6	1.908634	209.103
P7	2.188968	240.932
P8	3.754614	413.795
P9	2.294941	253.932
P10	2.372180	261.561
P11	2.855051	312.757
P12	2.402929	266.318

Minimum R: 1.544059° (~169.393 km)

Maximum R: 3.754614° (~413.795 km)

Mean R_prox: 2.237208°

Radius std: 0.566232°

2) Recalculated Epicenter (After Outlier Removal)

Epicenter (H, K): 10.126744° N, 124.632525° E

Epicenter 1σ interval: (H ± 1.181206°, K ± 5.360214°)

Distance to PHIVOLCS Measured Epicenter: 86.570 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P2	1.153182	128.014
P3	0.911752	101.382
P4	1.092552	121.262
P6	0.991852	110.001
P8	2.597805	287.241

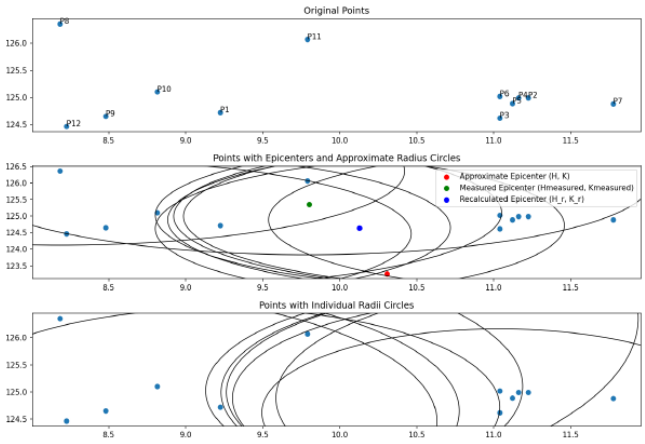
Minimum R: 0.911752° (~101.382 km)

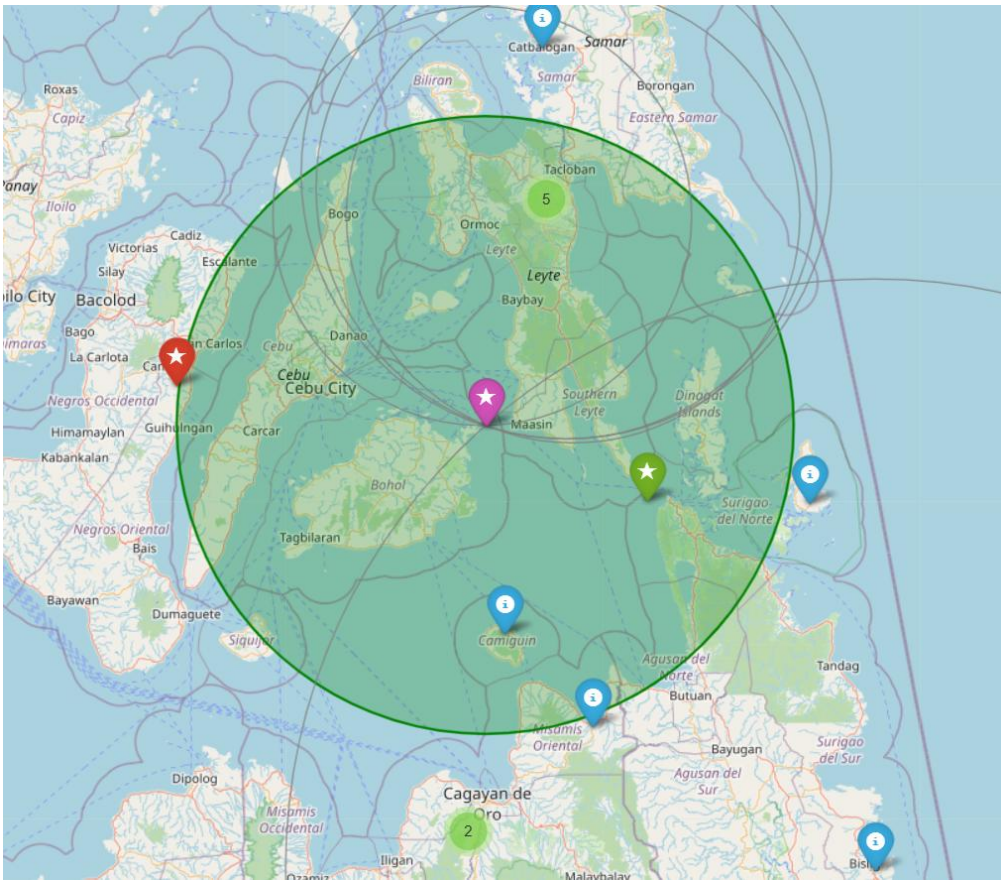
Maximum R: 2.597805° (~287.241 km)

Mean R_prox: 1.349428°

Radius std: 0.629661°

Figure: Triplet Geometry and Radii





Intensity II

Places	Coordinates	Findings (from the Method)
Cebu City, Cebu	10.2926° N, 123.9022° E	
Talacogon, Agusan del Sur	8.4480° N, 125.7864° E	
Dumaguete, Negros Oriental	9.2984° N, 123.2925° E	
Tagbiliran City, Bohol	9.6534° N, 123.8670° E	
Capoocan, Leyte	11.2709° N, 124.6193° E	
Carigara, Leyte	11.2591° N, 124.7078° E	
El Salvador City, Misamis Oriental	8.5599° N, 124.5269° E	
Oroquieta City, Misamis Occidental	8.4860° N, 123.7983° E	
Plaridel, Misamis Occidental	8.6115° N, 123.6959° E	

Intensity-Based Epicenter Estimate Report

1) Initial (Unfiltered) Epicenter Estimate

Epicenter (H, K): 9.100038° N, 122.099039° E
Epicenter 1σ interval: (H ± 4.792620°, K ± 26.198228°)
Distance to PHIVOLCS Measured Epicenter: 364.977 km

Radii to (H, K)

Point	Radius (deg)	Radius (km)
P1	2.161849	238.000
P2	3.744567	411.650
P3	1.209833	132.844
P4	1.852538	203.487
P5	3.326313	366.521
P6	3.386323	373.021
P7	2.487219	273.443
P8	1.806801	198.819
P9	1.669921	183.663

Minimum R: 1.209833° (~132.844 km)
Maximum R: 3.744567° (~411.650 km)
Mean R_prox: 2.405040°
Radius std: 0.836923°

2) Recalculated Epicenter (After Outlier Removal)

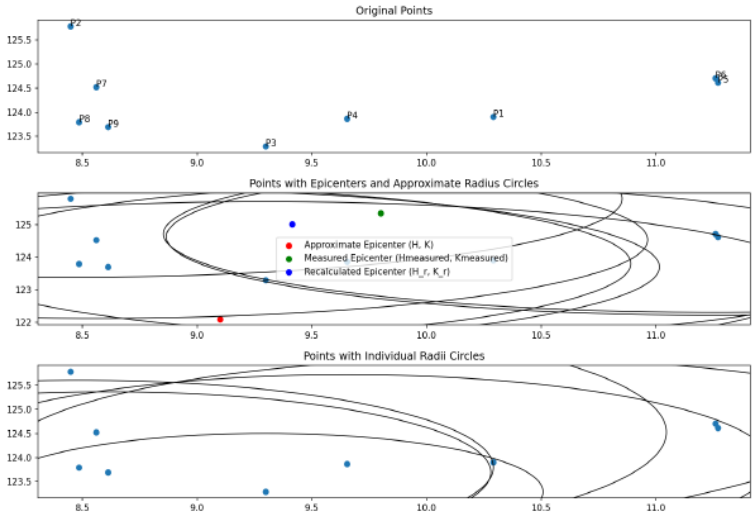
Epicenter (H, K): 9.414449° N, 125.006714° E
Epicenter 1σ interval: (H ± 1.280088°, K ± 1.964209°)
Distance to PHIVOLCS Measured Epicenter: 57.048 km

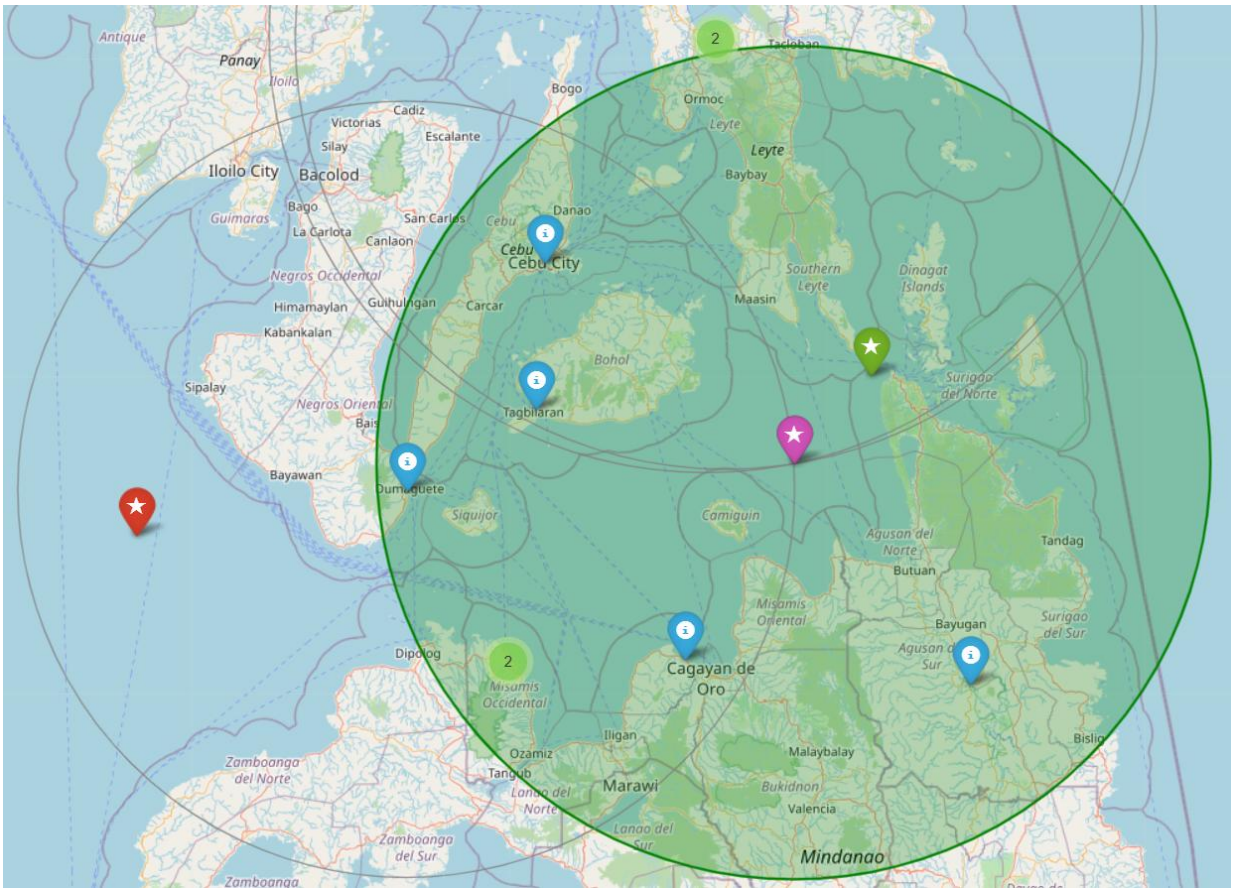
Radii to (H, K)

Point	Radius (deg)	Radius (km)
P3	1.718137	188.518
P5	1.896444	210.733
P6	1.868713	207.706

Minimum R: 1.718137° (~188.518 km)
Maximum R: 1.896444° (~210.733 km)
Mean R_prox: 1.827765°
Radius std: 0.078341°

Figure: Triplet Geometry and Radii





CONSOLIDATED TABLE				
Intensity	Initial Epicenter	Recalculated Epicenter	Outliers	Distance Error
Actual Location	9.80° N, 125.35° E	-	-	-
VIII	9.7613° N, 125.3968° E	9.7613° N, 125.3968° E	None	6.693 km
VI	9.5945° N, 125.5088° E	9.5945° N, 125.5088° E	None	28.719 km
V	10.2267° N, 125.8822° E	9.9853° N, 125.5583° E	Six (6) stable points: Liloan, Southern Leyte St. Bernard, Southern Leyte San Jose, Dinagat Islands Libjo, Dinagat Islands Cagdianao, Dinagat Islands	30.740 km
IV	9.1625° N, 122.2942° E	9.3990° N, 123.0615° E	Hinunangan, Southern Leyte Abugay, Leyte	254.836 km
III	10.3064° N, 123.2598° E	10.1267° N, 124.6325° E	Five (5) stable points: Tacloban City, Ormoc City, Palo, and Toloso, Leyte; Bislig City, Surigao del Sur	86.570 km
II	9.1000° N, 125.0990° E	9.4144° N, 125.0067° E	Two (2) stable points: Dumaguete, Negros Oriental, and Capoocan, Leyte	57.048 km

INTERPRETATION NARRATIVE BY INTENSITY LEVEL

(Instrumental epicenter: 9.80° N, 125.35° E).

INTENSITY VIII (Very Destructive)

Provinces covered: Surigao del Norte
Towns included: Ipil (Surigao City), Poblacion (San Francisco), Honrado (San Francisco), Macopa (San Francisco).

Interpretation
This dataset captures the immediate near-fault damage belt along the Surigao peninsula. Radii are tight and physically coherent because these sites sit close to the rupture trace, with steep bedrock and fault-zone amplification typical of the peninsula. The geometry offers good azimuthal spread (not purely collinear), yielding a stable solution.

After filtering / result

- **Recalculated Epicenter: 9.7613° N, 125.3968° E**
- **Distance from PHIVOLCS epicenter: 6.693 km**
- **Outliers:** None flagged in the consolidated table.

Assessment
As expected for severe near-source shaking, Intensity VIII provides the **best single-band constraint** on the epicentral region, with an error under ~7 km.

INTENSITY VI (Very Strong)

Provinces covered: Surigao del Norte
Towns included: Malimono, Mainit, Placer, Sison, Taganaan.

Interpretation
These sites outline the **strong-motion envelope** around the near-fault zone, spanning inland highlands and coastal settings with mixed substrates (volcanic ridges, metamorphic blocks, coastal plains). Site and path effects start to diversify the radii, increasing spread compared with VIII.

After filtering / result

- **Recalculated Epicenter: 9.5945° N, 125.5088° E**
- **Distance from PHIVOLCS epicenter: 28.719 km**
- **Outliers:** None flagged.

Assessment
Useful for defining the **broader strong-shaking belt**, but less precise for epicenter location than VIII—fully consistent with intensity–distance behavior.

INTENSITY V (Strong)

Provinces covered: Surigao del Norte, Southern Leyte, Dinagat Islands
Towns included (sample): Tubod, Bacuag, Alegria, Gigaquit, Claver (Surigao del Norte); Pintuyan, Libagon, St. Bernard, Liloan, San Francisco, San Ricardo (Southern Leyte); San Jose, Libjo, Cagdianao (Dinagat Islands).

Interpretation
This band spans **multiple islands** separated by short seas (Surigao Strait), demonstrating efficient lateral propagation of energy. Mixed geometry (multi-island, multi-direction) creates competing radius constraints; however, a subset of towns on **Dinagat + Southern Leyte** yields consistent bearings that stabilize the solution.

After filtering / result

- **Recalculated Epicenter: 9.9853° N, 125.5583° E**
- **Distance from PHIVOLCS epicenter: 30.740 km**
- **Stable points noted in your table:** Liloan & St. Bernard (Southern Leyte), San Jose–Libjo–Cagdianao (Dinagat).

Assessment
Still **operationally informative** for regional strong shaking, but inherently noisier than VIII–VI due to cross-island path and site effects.

INTENSITY IV (Moderately Strong)

Provinces covered: Southern Leyte, Leyte, Agusan del Norte
Towns included: Hinunangan, San Juan, Sogod, Macrohon, Maasin City (Southern Leyte); Abuyog, Mayorga (Leyte); Butuan City, Cabadbaran (Agusan del Norte).

Interpretation
These reports lie **farther from source** and include major sedimentary basins (e.g., Agusan plain), where basin amplification and long-period waves dominate over distance decay. The inversion tries to satisfy widely separated rings, leading to a large displacement.

After filtering / result

- **Recalculated Epicenter: 9.3990° N, 123.0615° E**
- **Distance from PHIVOLCS epicenter: 254.836 km**

- **Filtering notes (per table):** distant/basin-affected points drive the instability.

Assessment
Good for mapping **felt extent**, poor for locating the source. Large error is **physically expected** at this band.

INTENSITY III (Weak)

Provinces covered: Leyte, Samar, Surigao del Sur, Misamis Oriental, Lanao del Norte, Camiguin, Surigao del Norte
Towns included (sample): Tacloban, Ormoc, Palo, Pastrana, Tolosa, Catbalogan (Leyte/Samar); Bislig (Surigao del Sur); Cagayan de Oro, Gingoog (Misamis Oriental); Dapa (Siargao); Iligan (Lanao del Norte); Mambajao (Camiguin).

Interpretation
This halo spans **hundreds of kilometers**; low intensities reflect building response, perception, and local soils more than distance. After removing scattered outliers, a mid-range center emerges but remains far from the true epicenter.

After filtering / result

- **Recalculated Epicenter: 10.1267° N, 124.6325° E**
- **Distance from PHIVOLCS epicenter: 86.570 km**
- **Stable subset (per table):** Tacloban, Ormoc, Palo, Tolosa (Leyte); Bislig (Surigao del Sur).

Assessment
Valuable for **regional perceptibility** and attenuation trend checks; not reliable for pinpointing location.

INTENSITY II (Slightly Felt)

Provinces covered: Cebu, Agusan del Sur, Negros Oriental, Bohol, Leyte, Misamis Oriental, Misamis Occidental
Towns included: Cebu City; Talacogon; Dumaguete; Tagbilaran; Capoocan, Carigara; El Salvador City; Oroquieta City; Plaridel.

Interpretation
Reports define the **outermost perceptible fringe**. Readings are sparse and perception-dominated; any constraint is largely opportunistic, depending on geometry.

After filtering / result

- **Recalculated Epicenter: 9.4144° N, 125.0067° E**
- **Distance from PHIVOLCS epicenter: 57.048 km**
- **Stable pair (per table):** Dumaguete (Negros Oriental) and Capoocan (Leyte).

Assessment
Not suited for precise location; helpful to outline **maximum felt extent** and to corroborate decay trends.

CONSOLIDATED PHYSICAL INTERPRETATION

Macroseismic Field Behavior and Epicentral Region Characteristics

The Surigao 2017 field is **physically consistent** with a **shallow strike-slip rupture** on the Philippine Fault traversing the Surigao peninsula:

- **Highest Intensities (VIII): Near-Source Effects**
  Tightly grouped Surigao City–San Francisco localities identify the **damage core** and provide the best epicentral constraint (≈6.7 km).
- **Intermediate Intensities (VI–V): Main Strong-Motion Area**
  VI outlines a peninsula-wide ring; V crosses **short seas** into Dinagat and Southern Leyte, showing efficient inter-island propagation. A stable V-subset (Dinagat + S. Leyte) improves triangulation but remains moderately uncertain (~29–31 km).
- **Outer Intensities (IV–III–II): Regional Perceptibility Zone**
  These bands are **dominated by basin and site effects** (e.g., Agusan, Leyte). They map the felt extent but do not localize the source; IV shows the expected large displacement when distant basin towns are included.

Trends and Transitions Across Intensities

- **Spatial Contraction:** II–III form a wide multi-island halo; IV–V constrict toward northeastern Mindanao; VI–VIII collapse onto the Surigao peninsula around the fault zone.
- **Geological Coherence Increases with Intensity:** High intensities align with steep bedrock and rupture-adjacent ground; lower intensities reflect mixed soils and basin amplification.
- **Cross-Sea Propagation:** V demonstrates that **narrow straits do not block long-period energy**, only modulate it—hence strong shaking in Dinagat/Southern Leyte.
- **Distance-Error Progression:** Errors grow with decreasing intensity: VIII (~7 km) → VI/V (~29–31 km) → III/II (~57–87 km) → IV (≈255 km due to distant basin points).

Overall Physical Interpretation

The intensity pattern depicts a compact near-fault destruction zone, a broader but coherent strong-shaking belt that crosses to neighboring islands, and a wide regional felt area shaped by **basins, coastal plains, and inter-island paths**. This is the characteristic macroseismic signature of a shallow **Philippine Fault** earthquake in an archipelagic setting.